Von Neumann Computer Model

Q. Explain the concept of the Von Neumann Architecture. Discuss its components and working mechanism with the help of a block diagram.

> **Definition to Remember**

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1. The Stored-Program Concept

Before Von Neumann, early computers were "fixed-program" (physically rewired for new tasks). The Von Neumann concept revolutionized computing by storing instructions alongside data in memory, making machines universally programmable.

2. Three Main Components

1. **Central Processing Unit (CPU):** The brain, divided into:

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3. Block Diagram

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## 4. Working Mechanism (Fetch-Decode-Execute Cycle)

- **Fetch:** CU fetches the next instruction from memory using the Program Counter (PC).

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5. The Von Neumann Bottleneck

Because instructions and data share the same bus to memory, they must be fetched sequentially. Since CPUs are much faster than memory, the CPU wastes time idling. This limitation is known as the **Von Neumann Bottleneck**.

> **Must-Write Points (for 10 marks)**

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> **Quick Recall**

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